

Explanatory Note: Labour Market Outlook Dashboard

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Introduction

This dashboard visualises the evolution of labour market outcomes across EU member states. Utilising micro labour market data disseminated by [Eurostat](#), the statistical office of the European Union, it offers insights into the trends and changes in crucial labour market indicators.

Data

Main data sources

The dashboard features indicators derived from the harmonised EU Labour Force Survey (EU-LFS) micro data starting from 2006. It displays key labour market outcomes across gender and education groups. The indicators are calculated for the working age population, which covers individuals aged 15 to 64.

Secondary data sources

The generative AI exposure indicator is calculated using the occupational GenAI exposure scores developed by Gmyrek *et al* (2023).

Bruegel calculations

The Labour Market Outlook dashboard illustrates a diverse set of labour market indicators derived from harmonised EU-LFS. Alongside country-level data for EU member states, it includes EU average figures calculated using population weights. The dashboard goes beyond aggregate-level information, offering insights into gaps across gender and education groups when feasible. The dashboard highlights six demographic groups: i) low-educated men, ii) low-educated women, iii) medium-educated men, iv) medium-educated women, v) high-educated men and vi) high-educated women.

Table 1 lists the gender and education variables used to construct the six demographic groups. EU-LFS variable names are given in square brackets.

Table 1 Demographics

Variable [EU-LFS variable name]	
Gender [SEX]	Sex of the survey responded, coded as Male/Female.
Education [HATLEV1D]	The highest level of education successfully completed according to the International Standard Classification of Education (ISCED). ¹ •Low educated: at most lower secondary education, (ISCED 0-2) •Medium educated: upper secondary or post-secondary non-tertiary education, (ISCED3-4) •High educated: tertiary education, (ISCED5-8)
Weight [COEFFY]	All indicators are calculated using the population weights.

Labour market indicators

In the top-left section of the Labour Market Outlook dashboard, a summary of primary labour market indicators, namely labour force participation, employment and unemployment rates, is presented. This section of the dashboard also provides time series information on the evolution of those key labour market figures. Table 2 outlines the formulas used to construct these indicators.

Table 2 Labour market indicators

Variable [EU-LFS variable name]	
Labour force participation rate [ILOSTAT]	A measure of the proportion of a country's working-age population that engages actively in the labour market, either by working or looking for work. $LFPR = \frac{Employed + Unemployed}{Working\ age\ population}$
Employment rate [ILOSTAT]	A measure of the extent to which available labour resources (people available to work) are being used. $Employed = \frac{Employed}{Working\ age\ population}$
Unemployment rate [ILOSTAT]	The percentage of people in the labour force who are unemployed. $Unemployed = \frac{Unemployed}{Labour\ force}$
Weight [COEFFY]	All indicators are calculated using the population weights.

Job characteristics

The lower-left section of the Labour Market Outlook dashboard depicts diverse job characteristics through four bar charts. The bold figures in the top-left corner of these charts indicate country

¹ ISCED levels correspond to:

ISCED level 0 – Early childhood education

ISCED level 1 – Primary education

ISCED level 2 – Lower secondary education

ISCED level 3 – Upper secondary education

ISCED level 4 – Post-secondary non-tertiary education

ISCED level 5 – Short-cycle tertiary education

ISCED level 6 – Bachelor's or equivalent level

ISCED level 7 – Master's or equivalent level

ISCED level 8 – Doctoral or equivalent level

For more information on ISCED classification please refer to [Unesco](#).

averages, with the bars representing the six demographic groups. Table 3 explains how the four job characteristics measures are constructed.

Table 3 Job characteristics

Variable [EU-LFS variable name]	Definition
Temporary jobs [TEMP]	The percentage of employees with a temporary contract. $\text{Temporary} = \frac{\text{Nb persons with temporary contracts}}{\text{Employed}}$
Part-time jobs [FTPT]	Percentage of people who have a part-time job. $\text{Part time} = \frac{\text{Nb persons with part - time job}}{\text{Employed}}$
Working from home [HOMEWORK]	Percentage of people who sometimes or mainly works from home. $\text{Home} = \frac{\text{Nb persons (partly) working from home}}{\text{Employed}}$
GenAI exposure [based on Gmyrek et al (2023)]	Generative AI exposure by occupation. Computed using the occupation specific GenAI exposure scores developed by Gmyrek et al (2023). Tasks are rated on a scale of 0 to 1, where values closer to 1 indicate higher susceptibility to automation. The sum and standard deviation of the tasks within a specific occupation determines whether the occupation is categorized as having potential to be augmented or automated, or to lie at the junction of two (“the big unknown”). For more detail see Gmyrek et al. (2023).
Weight [COEFFY]	All indicators are calculated using the population weights.

Working age population

In the upper-right section of the Labour Market Outlook dashboard, population figures are presented. The table distinguishes between the three main labour market statuses and offers information on structural labour market issues, such as long-term unemployment or individuals who are willing to work but are not actively seeking employment due to health reasons, discouragement, or caregiving responsibilities.

Table 4 lists EU-LFS variables that are used to construct working age population series.

Table 4 Working age population, age 15 to 64

Variable [EU-LFS variable name]	
Working age population [AGE_GRP]	Population aged 15 to 64.
Employed [ILOSTAT]	Total number of people who are employed (ILOSTAT=1)
Unemployed [ILOSTAT]	Total number of people who are unemployed (ILOSTAT=2)
Inactive [ILOSTAT]	Total number of people who are inactive (ILOSTAT=3)
Long-term unemployed [DURUNE]	Number of people who have been looking for employment for more than a year. (DURUNE=3 and DURUNE=4)
Inactive, wants to work [ILOSTAT, WANTWORK]	Total number of people who are inactive (ILOSTAT=3) but willing to work (WANTWORK=1)
Health [SEEKREAS]	Number of people who are willing to work but do not actively seek employment due to own illness or disability (SEEKREAS=3)

Discouraged [SEEKREAS]	Number of people who are willing to work but do not actively seek employment due to the belief that no suitable job is available (SEEKREAS=1)
Care [SEEKREAS]	Number of people who are willing to work but do not actively seek employment due to care responsibilities (SEEKREAS=4)
Weight [COEFFY]	All indicators are calculated using the population weights.

Indicators by country

The heat map uses darker colours to represent higher parameter values and illustrates cross country variation in main labour market indicators. Table 5 lists these indicators along with the EU-LFS variables used for their computation. The first column presents the indicators, with EU-LFS variable names given in square brackets, while the second column outlines the methodology behind the computation of these indicators.

Table 5 Indicators by country

Variable [EU-LFS variable name]	
Labour force participation rate [ILOSTAT]	A measure of the proportion of a country's working-age population that engages actively in the labour market, either by working or looking for work. $LFPR = \frac{Employed + Unemployed}{Working\ age\ population}$
Employment rate [ILOSTAT]	A measure of the extent to which available labour resources (people available to work) are being used. $Employed = \frac{Employed}{Working\ age\ population}$
Unemployment rate [ILOSTAT]	The percentage of people in the labour force who are unemployed. $Unemployed = \frac{Unemployed}{Labour\ force}$
Long-term unemployed [DURUNE]	The percentage of people in the labour force who are unemployed and who have been looking for employment for a year. $Long\ term = \frac{LT_unemployed}{Unemployed}$
Inactivity rate [ILOSTAT]	Is the proportion of people outside the labour force (i.e. economically inactive persons). $Inactive = \frac{Inactive}{Working\ age\ population}$
Health [SEEKREAS]	Main reason for not searching for employment: Own illness or disability (SEEKREAS=3) $Health = \frac{\#\ of\ SEEKREAS = 3}{Inactive, but wants to work}$
Discouraged [SEEKREAS]	Main reason for not searching for employment: No suitable job is available (SEEKREAS=1) $Discouraged = \frac{\#\ of\ SEEKREAS = 1}{Inactive, but wants to work}$
Care [SEEKREAS]	Main reason for not searching for employment: Care responsibilities (SEEKREAS=4) $Care = \frac{\#\ of\ SEEKREAS = 4}{Inactive, but wants to work}$
Temporary [TEMP]	The percentage of workers with a temporary contract $Temporary = \frac{\#\ of\ people\ with\ temporary\ contracts}{Employed}$
Part-time [FTPT]	Percentage of people among employed that have a part-time job.

	Part – time = $\frac{\# \text{ of people with part – time job}}{\text{Employed}}$
Home [HOMEWORK]	Percentage of people among employed that sometimes or mainly works from home. Home = $\frac{\# \text{ of people (partly) working from home}}{\text{Employed}}$
Atypical [SHIFTWK, EVENWK, NIGHTWK, SATWK, SUNWK]	The proportion of individuals in employment that are engaged in either shift work, evening employment, night shifts or weekend shifts. Atypical = $\frac{\# \text{ of people with atypical shifts}}{\text{Employed}}$
GenAI exposure [based on Gmyrek <i>et al</i> (2023)]	Generative AI exposure by occupation. Computed using the occupation specific GenAI exposure scores developed by Gmyrek <i>et al</i> (2023). Tasks are rated on a scale of 0 to 1, where values closer to 1 indicate higher susceptibility to automation. The sum and standard deviation of the tasks within a specific occupation determines whether the occupation is categorized as having potential to be augmented or automated, or to lie at the junction of two (“the big unknown”). For more detail see Gmyrek <i>et al</i> . (2023).
Training rate [EDUCNFE4]	Participation in non-formal education and training in the last 4 weeks. Training = $\frac{\# \text{ of people who participated training}}{\text{Employed}}$
Inactive, wants to work [ILOSTAT, WANTWORK]	Total share of people who are inactive but wants to work (ILOSTAT=3, WANTWORK=1), in total inactives (ILOSTAT=3). Wants to work = $\frac{\# \text{ of people who are inactive but wants to work}}{\text{Inactive}}$
Weight [COEFFY]	All indicators are calculated using the population weights.

Gap by gender

The gap in the labour market indicators by gender is calculated as a simple subtraction. For a given indicator, x , the gap equals to the difference between the values of men and women.

$$xGap_{gender} = x_{men} - x_{women}$$

A larger absolute value indicates a more substantial difference between men and women for the specified indicator. A negative value signifies that the indicator values for women exceed those of men. The measurements are expressed in percentage points (pp).

Example: In Italy in 2022, male labour force participation rate was 74.6%, while female labour force participation rate was only 56.4%. The labour force participation rate of men exceeded that of women by **18.2pp** (Figure 1, left panel).

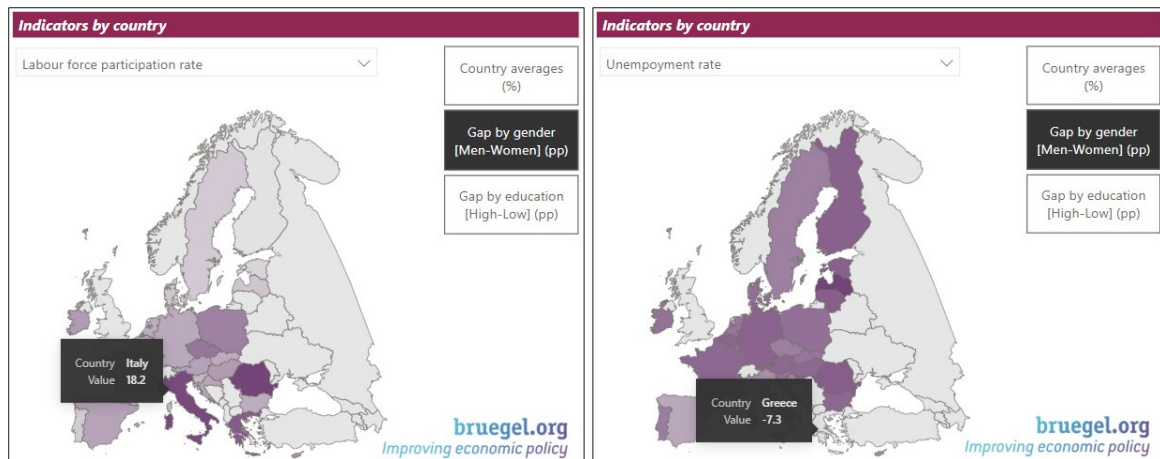


Figure 1 Gap by gender

In the same year in Greece, male unemployment was 9.3% where female unemployment rate was as high as 16.6%. Male unemployment rate was **7.3pp lower** than female unemployment rate (Figure 1, right panel).

Gap by education

Similarly, the gap by education level is determined through a straightforward subtraction process.

$$xGap_{educ} = x_{high} - x_{low}$$

A larger absolute value indicates a more significant difference between high and low educated individuals for the specified indicator. A negative value signifies that the indicator values for low-educated individuals exceed those with high education levels. The measurements are expressed in percentage points (pp).

Example: In the year 2022 in Poland, the labour force participation rate was 91.5% for high educated people while it was only 26.3% for their low educated counterparts. Individuals with high education exhibited a labour force participation rate that exceeded those with low levels of education by **65.2pp** (Figure 2, left panel).

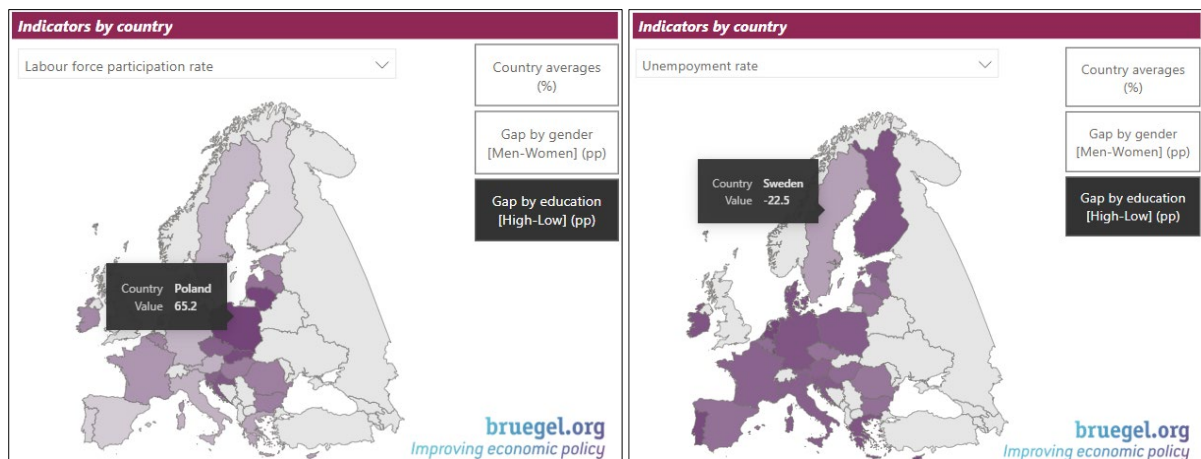


Figure 2 Gap by education

In the same year in Sweden, unemployment rate was 25.8% for low educated people while it was only 3.3% for people with high education. The unemployment rate of individuals with high education levels was **22.5pp lower** than their low educated counterparts (Figure 2, right panel).

Notes on data

EU-LFS confidentiality and data reliability

- The EU-LFS, like all surveys, is based on a sample of the population and is therefore subject to typical sampling errors. To ensure the statistical reliability of published results and protect respondent confidentiality, Eurostat follows specific guidelines.
- One such measure is the **Confidentiality Threshold**, which prevents the publication of cells with fewer than three observations in the unweighted sample. These cells are excluded from the Labour Market Outlook Dashboard statistics.
- In addition, data reliability is maintained through **Reliability Thresholds**. Eurostat sets two limits, referred to as 'a' and 'b', which depend on the sample size and design in each Member State. These limits, expressed in thousands of the weighted population, vary by country, year, dataset type (quarterly/yearly), and subsample.

According to Eurostat guidelines, the Labour Market Outlook Dashboard does not publish estimates for (weighted) populations below limit 'a'. However, estimates falling between limits 'a' and 'b' are published. Hence the user is advised to refer to Table 8 in the Appendix which displays limit b values for 27 member states for 2006-2023 period.

The specific values for thresholds 'a' and 'b' are provided in an [Excel dataset](#) on the Eurostat website. Further details can also be found in the [LFS Statistics Explained](#) pages under the section "Publication guidelines and thresholds".

EU-LFS data comparability

- Enhanced harmonization, improved content stability, and increased survey frequency enables comparability among successive EU-LFS surveys. However, adjustments to population data following a census, revised classifications/regulations, and other country-level modifications aimed at improving survey quality may have implications for comparability over time.

For example, the 2011 Census required updates to population figures in several countries, while the 2016 Census in **Ireland** resulted in a break in all series in 2017.

More information on historical breaks can e.g. be found at [EU LFS- Comparability over time and across countries](#).

- **In 2014**, the classification of educational activities changed from ISCED 1997 and to **ISCED 2011**. For the aggregated education levels used in this dashboard (low – medium – high level of education) data are comparable over time for all available countries except Austria and Estonia.

Austria: In ISCED 2011, the qualification obtained upon completing higher technical and vocational colleges is categorized under ISCED level 5. Previously, under ISCED 1997, the same qualification was categorized under ISCED level 4 but earmarked as equivalent to tertiary education.

Estonia: the level shift is due to the reclassification of a program with durations ranging from 6 months to 3.5 years: ISCED 2011 classifies the qualification obtained upon successful completion of "vocational courses based on basic education" under ISCED level 2. This degree was reported as ISCED level 3 in ISCED 1997.

- Since **2021**, the EU-LFS is based on Regulation (EU) 2019/1700, also called the Integrated European Social Statistics Framework Regulation (**IESS FR**), and its Commission Implementing Regulation (EU) 2019/2240. A general overview of the changes to the EU-LFS in 2021 is given in chapter 3 of the [2021 EU LFS User Guide](#).

To enable overtime comparisons, Eurostat transcoded the datasets preceding 2021 as far as possible to match the variables according to the IESS coding scheme.

- To improve survey quality, countries occasionally make alterations in the survey implementation. These alterations may involve changes to survey or sample design, weighting scheme, or questionnaire content.

Table 6 lists these type of country specific breaks which took place from 2006 to 2023, (b) donates breaks in the series while (d) indicates a change in definitions. (b)^c represents years where the break is due to census revisions. Years 2014 and 2021 are marked with an asterisk in Table 6, to indicate the switch to **ISCED 2011** in the classification of educational attainment (2014) and the implementation of **IESS FR** (2021).

Table 6 EU-LFS breaks

	2006	2007	2008	2009	2010	2011	2012	2013	2014*	2015	2016	2017	2018	2019	2020	2021*	2022	2023
AT		(b)							(b)							(b)		
BE			(b)			(b)			(b)			(b)				(b)		
BG			(b)		(b) ^c	(b)			(b)							(b)		
CY				(b) ^c					(b)							(b)		
CZ						(b) ^c			(b)							(b)		
DE					(b) ^c	(b)	(b)		(b)						(b)	(b)		
DK		(b)	(b)						(b)		(b)	(b)				(b)		
EE									(b)							(b)		
EL				(b)					(b)							(b)		
ES									(b)							(bd)	(d)	
FI		(b)	(b)					(b)	(b)							(b)		
FR									(b)							(bd)	(d)	
HR					(b) ^c				(b)							(b)		
HU									(b)							(b)		
IE		(b) ^c							(b)			(b)				(b)		
IT									(b)							(b)		
LT									(b)							(b)		
LU		(b)		(b)					(b)	(b)						(b)		
LV									(b)							(b)		
MT						(b)			(b)							(b)		
NL								(b)	(b)					(b)		(b)		
PL					(b) ^c				(b)							(b)		
PT						(b)			(b)							(b)		
RO					(b)				(b)							(b)		
SE									(b)				(b)			(b)		
SI									(b)							(b)		
SK						(b) ^c			(b)							(b)		

- The breaks which might have implications for the variables used in the development of Labour market outlook dashboard are listed in Table 7 in the Appendix. Table 7 presents information extracted from a Eurostat document detailing specific breaks for each country. For a comprehensive list, please consult: [Country specific breaks.xls](#).

- Core country questionnaires are available at [EU LFS - Core questionnaires](#).
- For each reference year, LFS quality reports are published at [LFS Quality](#).

Notes on Labour market outlook dashboard data

- Starting from 2021, variables used in the computation of atypical work indicator are collected only in odd years. Hence this indicator is missing in 2022.
- In 2011, there was a break in the ISCO classification (from ISCO88 to ISCO08). This re-classification of occupations renders it impossible to make meaningful comparisons of occupations before and after 2010, therefore the GenAI exposure analysis starts in 2011. Moreover, Bulgaria, Slovenia and Malta are omitted from this analysis as they do not have the occupation information at the three-digit level.
- Training participation variable [EDUCNFE4] only covers the four weeks preceding the survey. Hence, the participation rates calculated using this variable are considerably lower compared to the training participation rates derived from other surveys such as Adult Education Survey, where respondents are prompted to recall their training participation over the past year.

Country specific data issues

- **Belgium:** SEEKREAS variable is missing in 2011, hence *Care*, *Health* and *Discouraged* cannot be calculated from the survey data. Nevertheless, a proxy figure, the simple average of 2010-2012 numbers are still displayed in the Working age population table.

Similarly, SEEKREAS variable is missing in 2020 in **Germany**, and **Care**, **Health** and **Discouraged** cannot be calculated using the survey data. We again used the proxy figure, the simple average of 2019-2021, and displayed these numbers in the Working age population table.

Main motivation of this was to prevent a misinterpretation of the 2020 data across the EU. 2020 is particularly crucial for assessing the number of individuals inactive due to health reasons, given the significant impact of the Covid-19 pandemic, which led to a surge in these figures across all EU member states. However, the absence of data from Germany was creating a false impression that the number of individuals inactive due to health reasons decreased at the EU level in 2020. By substituting proxy figures for Germany, this problem is partially addressed, albeit at the cost of spreading out the 2020 shock over a three-year period (2019-2021).

- **Croatia:** Population figures based on last census results show a drop in the population in 2022. LFS data weighting for 2021 is still based on previous population figures hence the size of working age population gathered from the EU-LFS is bigger than the actual figures. The survey weights are expected to be adjusted in the future.
- **Cyprus:** LFS data covers only the population in the territory controlled by the government of the Republic of Cyprus. Hence, the working age population is lower than the working age population in the whole island.
- **Finland:** SEEKREAS variable is partially missing in 2006 and 2007, hence **Care** cannot be calculated.
- **Italy:** The number of people who are **Discouraged** decreases drastically from 2020 to 2021.
- **Luxembourg:** SEEKREAS variable is partially missing in 2011, hence **Discouraged** cannot be calculated.
- **Malta:** EU-LFS is available only 2009 onwards.

- **Sweden:** DURUNE variable is missing in 2006, hence **Long-term unemployed** cannot be calculated.

References

Gmyrek, P., J. Berg and D. Bescond (2023) 'Generative AI and jobs: A global analysis of potential effects on job quantity and quality', *ILO Working Paper 96*, International Labour Organization, available at <https://doi.org/10.54394/FHEM8239>

Appendix

Table 7 List of EU-LFS breaks by country

Country	Reason	Variable(s) affected	Year
AT	Break in series due to the reclassification of a programme spanning levels: The qualification acquired upon successful completion of higher technical and vocational colleges is allocated in ISCED 2011 to ISCED level 5; under ISCED 1997 the same qualification was reported on ISCED level 4, but earmarked as equivalent to tertiary education.	EDUCLEVEL HAT11LEV	2014
BE	New national questions to measure level of education.	HAT11LEV, EDUCLEVEL	2014
BE	Panel design, wave approach, mixed mode data collection and a revision of calibration method + introduction of nonresponse bias adjustment	ALL	2017
BE	Change in the editing procedures among the 15-17 year old who claim not wanting to work because in education + declaring to be a student in the MAINSTAT variable, while not reporting any participation in education on EDUCFED4	EDUCFED4, EDUC4WEEKS	2021
BG	Changes in weighting procedure: - calculation of weighting factor by 5-years age group - excluding from the population data people living in institutional households	ALL	2011
CH	Weight revision due to inclusion of administrative data	ALL	2010
CZ	Improvements of the classification of occupations (ISCO4D - first digit) through training for the interviewers and the check of reports: The decline, in comparison the year 2009 and 2010, was about 60 thousands of employed person in main class 2 (highly skilled non-manual occupations) and rise in major groups 3 to 5 (low skilled non-manual occupations).	ISCO	2010
CZ	Revision of the variable COURATT (attendance of taught learning activities - non formal education: The revised questionnaire only for 4th quarter of 2011 as a part of Pilot "Revision of education variables in the EU LFS", but with impact for all following year 2012 (repeated visits due to the rotation scheme of consecutive 5 quarters).	COURATT, EDUC4WN	2011
CZ	Revision of the variable COURATT (attendance of taught learning activities - non formal education: The revised questionnaire only for 4th quarter of 2011 as a part of Pilot "Revision of education variables in the EU LFS", but with impact for all following year 2012 (repeated visits due to the rotation scheme of consecutive 5 quarters).	COURATT, EDUC4WN	2013
DE	Complete redesign of questionnaire (leading questions on employment).	ILOSTAT	2011
DE	Change of the national classification of occupation (KldB92 -> KldB2010), including some revisions for 2012. The classification 2010 is used to code ISCO 08.	ISCO	2012
DE	New micro census from 2020: structural changes plus implementation issues combined with COVID-19 effects	ALL	2020
DK	New survey structure (including significant increase of sample size and changes in auxiliary information) with impact e.g. on education data.	HAT97LEV HATFIELD HATYEAR EDUCLEVEL COURATT EDUC4WN EVENWK NIGHTWK SATWK SUNWK	2007
DK	Weight revision	ALL	2008
DK	Survey changes (data collector, move to CAWI, low response rate)	ALL	2016
DK	Change of data collector, increased response rate	ALL	2017
EE	Break in series due to the reclassification of the programme 'vocational courses based on basic education' that is allocated to ISCED level 2 under ISCED 2011 while under ISCED 1997 it was classified as ISCED level 3.	EDUCLEVEL HAT11LEV	2014
EL	Question on participation in non-formal education changed from a simple yes no question to a question with many answer categories. While in 2007 the mean participation rate in non formal education was 2.5%, in 2008 rose to 3.5%	COURATT, EDUC4WN	2008

Table 7 List of EU-LFS breaks by country (cont'd)

Country	Reason	Variable(s) affected	Year
EL	New primary sampling units were selected and new sampling frames were compiled for the 4/6 of the sample. The update of the sample will be completed by the 2nd quarter 2010.	ALL	2009
ES	National version of the ISCO-08.COM was implemented in the Spanish EPA in 2011.	ISCO	2011
ES	Change in the classification of non-formal education activities.	COURATT, EDUC4WN	2014
FI	Changes in the variable TEMPDUR: The variables refers to the duration of the current contract and not to the total time working continuously in the same job.	TEMPDUR	2010
FR	Education and training variables are affected by changes in the questionnaire and the survey process: the attained education level is affected from 2013 Q1 to 2014 Q2, due to the rotational pattern; the break regarding access to lifelong learning appears in 2013 Q1 only. NOTE: previous breaks in the ILO labour status in 2013 Q1 have been removed by back-data weight revisions disseminated since autumn 2015.	EDUCSTAT, EDUCLEVL, EDUCFILD, COURATT, COURLEN, COURPURP, COURFILD, COURWORH, HATVOC, HATFIELD, HAT97LEV, EDUC4WN, HATYEAR	2013
FR	Introduction of a quarterly continuous survey in FR overseas departments and inclusion of data for FR overseas departments in quarterly datasets.	ALL	2014
HR	Revision of the variables HAT97LEV and EDUCLEVEL in 2010 (new answer options added in the national questionnaire).	HAT97LEV, EDUCLEVEL	2010
HU	For the variable COURATT additional clarifications and reminders were added in order to better cover compulsory training systems and introduction courses for those who started their job recently	COURATT, EDUC4WN	2015
IE	Shift from seasonal to calendar quarter	ALL	2007
IE	Prior to Q2 2012 the SEEKREAS question was only asked of respondents on our side who had answered YES to WANTWORK.	SEEKREAS	2012
IE	Survey break: From 2017 Q3 onwards, the Irish LFS is based on a new survey (introduction of a new sample, introduction of CATI, redesigned questionnaire, data processing changes, and other methodological enhancements). A micro data weight revision corrects for breaks in headline indicators on labour status by age and economic sector back to 2007. The series for other indicators before and after the introduction of the new survey may not be directly comparable.	ALL	2017
IS	Changed weighting scheme	ALL	2020
LU	Re-organisation of fieldwork and sampling with impact on survey results and inclusion of students in holidays in the variable EDUCSTAT	ALL	2009
LU	Special yearly weights	ALL (Yearly)	2009
LU	Introduction of infra-annual rotational pattern	ALL	2015
LV	Change in the variables NATIONAL: A further code NN was introduced in Europe to distinguish citizens and recognized non-citizens (holders of non-citizen passports existing e.g. in Latvia – code also retroactively used from 2008 onwards).	NATIONAL	2008
MT	Different treatment of responses based on dependent interviewing.	SEEKDUR*	2010
MT	Changes in questionnaire in Q1 2011 to collect education variables.	HAT97LEV, EDUCLEVL	2011
MT	Changes in the questionnaire related to the variable COURATT.	COURATT	2017
NL	New data collection mode (web) - start 2012Q4 (only 1 out of 5 waves but in Q1 2013 4 out of 5 waves, thus possible breaks will be more profound.	NACE, ISCO, HATLEVEL, EDUCLEVL, HATFIELD	2013

Table 7 List of EU-LFS breaks by country (cont'd)

Country	Reason	Variable(s) affected	Year
NL	Improvement in coding for people with graduation outside the Netherlands	HAT11LEV, HATYEAR, HATVOC, HATFIELD	2019
PL	Since the first quarter of 2008 there has been introduced a change in the methodology of calculating a duration of search for employment (SEEKDUR). A question concerning a break in job search of at least 4 weeks duration has been introduced into the questionnaire. The duration of job search is counted from the moment of the end of the break in job search (if there was such break). Therefore, these data are not comparable to the data for the previous quarters. For the variable METHODM - other method used, there were specified other active methods and other inactive methods. Since the first quarter of 2008, in the variable METHODM have been coded only other active methods. YEARESID - The coding of the variable YEARESID years of residence in this country has been adjusted to the change described in the Regulation No 102/2007 of 2 February 2007 EDUCSTAT - The EDUCSTAT variable was supplemented with code "person in regular education but on holidays". This code in case of Poland concerns mainly persons who completed one level of education (school, university) and are waiting for the results of recruitment into the next level of education (school, university). New NACE Rev. 2 classification	SEEKDUR* METHODM YEARESID EDUCSTAT EDUC4WN NACE	2008
PL	Increase of the sample size (100%) Generalization of results over the overall population carried out with the use of the data on population of Poland aged 15 years and more coming from the balances compiled on the basis of the Population and Census 2011 and together with the methodological changes targeted at harmonization of the population covered with the survey in accordance with Eurostat recommendations (persons absent from a household for 12 months or longer are excluded from the survey range - before the absence was to be over 3 months). EDUCSTAT, COURATT, HATFIELD, HATYEAR: Covers also people aged 65 and more.	ALL EDUCSTAT, COURATT, EDUC4WN, HATFIELD, HATYEAR	2010
PL	New ISCO-08 classification	ISCO	2011
PL	For variable COURATT <i>"Did you attend any courses, seminars, conferences or receive private lessons or instructions outside the regular education system (hereafter mentioned as taught learning activities) within the last four weeks?" the following clarification was introduced into the questionnaire "there should be also included courses related to personal intellectual or physical development or one's own interest, e.g. dancing course, swimming course, cookery course, judo, karate, driving course, etc." Previously this information was only included in "Explanatory Notes to the Labour Force Survey"</i>	COURATT, EDUC4WN	2013
PL	Questionnaire changes to allow better coverage of participation in non-formal education	COURATT, EDUC4WN	2018
PT	A new data collection methodology for the Labour Force Survey was implemented. This change involved a break in the series for Portuguese LFS data. In particular the unemployment rate for Portugal according to the new methodology was 12.4% in the 1st quarter of 2011 while, if the previous collection mode had been continued, this rate would have been 11.4%.	ALL	2011
RS	Sample revision including introduction of rotational pattern, new weighting procedure	ALL	2014
SE	Changes were made for the variable COURATT so that it now also includes informal education during working hours in the main job and the 4 week reference period was introduced for COURATT in Q1 2008.	COURATT, EDUC4WN	2008
SE	Statistics Sweden revised LFS statistics from 2018 Q3 onwards, using only half of the sample, due to an earlier identification of quality deficiencies. Revised results are based on data collection carried out by Statistics Sweden's Data Collection Department only. However, as the statistics are now based on half of the sample, this increases the uncertainty, particularly at a more disaggregated level. For more information see: https://www.scb.se/en/finding-statistics/statistics-by-subject-area/labour-market/labour-force-surveys/labour-force-surveys-lfs/pong/statistical-news/labour-force-surveys-lfs-october-2019/	ALL	2018

* SEEKDUR variable is used in the construction of DURUNE variable.

Table 8 Reliability thresholds (limit b)

Year	AT	BE	BG	CY	CZ	DE	DK	EE	EL	ES	FI	FR	HR	HU	IE	IT	LT	LU	LV	MT	NL	PL	PT	RO	SE	SI	SK
2006	6	5	8	1.5	3	50	4	3.8	3.5	8	4	50	24	4.8	4.1	2.5	22.5	1	3.3	0.9	5	15	4.5	11.5	3.5	4	3
2007	6	5	9	1.5	3	50	4	3	3.5	8	4	50	10	4.8	3.1	2.5	22.5	1	1.6	0.9	5	15	4.5	11.5	3.5	4	3
2008	6	5	12	1.5	3	50	4	2.95	3.5	8	4	50	10	4.8	3.1	2.5	22.5	1	1.8	0.75	5	15	4.5	11.5	3.5	4	3
2009	6	5	12	1.5	3	50	4	3.13	3.5	8	4	44	10	4.8	3.1	2.5	8	1	2.7	0.75	5	15	4.5	11.5	3.5	4	3
2010	6	5	14	1.5	3	50	4	3.05	3.5	8	4	40	10	4.8	3.1	2.5	8	1	3.6	0.75	6.5	15	4.5	11.5	2	4	3
2011	6	5	14	1.5	6	50	4	2.6	3.5	8	4	40	10	4.8	3.1	2.5	8	1	3.1	0.75	6.5	15	4.5	11.5	2	4	3
2012	6	5	16	1.5	6	5	4	2.6	3.5	8	4	40	10	4.8	3.1	2.5	8	1	2.6	0.8	6.5	15	4.5	11.5	2	4	3
2013	6	5	12.6	1.5	6	5	4	2.6	3.5	8	4	40	10	4.8	3.1	2.5	4.5	1	2.7	0.85	6.5	15	4.5	11.5	2	4	3
2014	6	5	12	1.5	6	5	4	2.5	3.5	8	4	40	10	4.8	3.1	2.5	4.5	1	2.3	0.85	6.5	15	4.5	11.5	2	4	3
2015	6	5	13.9	1.5	6	5	4	2.5	3.5	8	4	40	10	4.8	3.1	2.5	4.5	1	2	0.8	6.5	15	4.5	11.5	2	4	3
2016	6	5	11.5	1.5	6	5	4	2.5	3.5	8	4	40	10	4.8	3	2.5	4	1	2	0.85	6.5	15	4.5	11.5	2	4	3
2017	6	8	12.3	1.5	6	5	4	2.5	3.5	8	4	40	10	4.8	3	2.5	4	1	1.5	0.9	6.5	15	4.5	11.5	2	4	3
2018	6	8	10.1	1.5	6	5	4	2.4	3.5	8	4	40	10	4.8	3	2.5	4	1	2.4	0.9	6.5	15	4.5	11.5	3	4	4
2019	6	8	11.7	1.5	6	5	4	2.4	3.5	8	4	40	10	4.8	6.6	2.5	4	1	2.1	1.1	6.5	15	4.5	11.5	6.4	4	4
2020	6	8	15.3	1.5	6	40	4	2.4	3.5	8	4	40	10	4.8	7.8	2.5	4	1	2.3	1.15	6.5	20	4.5	11.5	6.4	4	4
2021	6	8	12.2	1.5	6	50	4	4.6	8.5	8	4	40	10	4.8	11.4	3.5	4	1	7.8	2.2	6.5	20	8.6	11.5	6.4	5	10.5
2022	14	8	11.3	1.5	6	50	4	2.2	8.5	8	4	40	10	4.8	11.5	8.5	4	1	7.9	2.5	6.5	20	8.6	11.5	6.4	12	9
2023	14	8	11.8	1.5	6	50	4	2.6	8.5	8	7	40	10	4.8	11.5	8.5	4	4	9	2.5	6.5	20	8.7	11.5	6.4	12	7.5